

Multi-Member Districts and Proportional Representation: Exploring Alternatives to Single-Member Redistricting

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Abstract

1 Introduction

Congressional redistricting in the United States operates under a fundamental constraint: single-member districts elected by winner-take-all plurality voting. This combination creates structural incentives for gerrymandering while systematically excluding minor parties from representation [? ?]. The 435 single-member districts that constitute the U.S. House of Representatives ensure that even when minor parties (Libertarian, Green, etc.) receive 3-5% of the national vote, they win zero seats [?]. Meanwhile, the winner-take-all mechanism enables partisan gerrymandering through the strategic manipulation of district boundaries—packing opposition voters into supermajority districts while cracking remaining voters across districts where they form ineffective minorities [2].

This institutional design is not universal. Among established democracies, the United States is exceptional in its exclusive reliance on single-member districts for legislative elections [?]. Most parliamentary democracies employ multi-member districts (MMDs) with proportional representation (PR), allocating multiple seats per district in proportion to each party’s vote share [?]. Ireland uses 3-5 member constituencies with the single transferable vote [?]. Germany’s mixed-member proportional system combines single-member districts with party-list seats to ensure nationwide proportionality [?]. New Zealand switched from first-past-the-post to mixed-member proportional representation in 1996 following a citizen referendum, motivated by concerns about disproportional representation and wasted votes [?].

The United States has historical experience with multi-member districts. From 1789 to 1842, many states elected representatives through at-large or multi-member constituency elections [?]. The Apportionment Act of 1842 first required single-member districts, though enforcement was inconsistent. The Uniform Congressional District Act of 1967 codified the single-member requirement in federal law [?]. This institutional choice was not constitutionally mandated—the Constitution is silent on district magnitude—but rather reflects

legislative preference for geographically localized representation and incumbent protection through the two-party system [?].

This paper explores algorithmic redistricting with multi-member districts as an alternative to the current system. We design three MMD configurations for the U.S. House of Representatives using recursive bisection graph partitioning: 145 districts of 3 representatives each, 109 districts of 4 representatives, and 87 districts of 5 representatives (totaling 435 seats). For each configuration, we simulate the 2020 presidential election using party-list proportional representation with the D’Hondt seat allocation method [?]. Our analysis quantifies four dimensions: proportionality (Gallagher index), minor party representation (seat shares for parties receiving <5% national vote), minority representation (comparing proportional outcomes to single-member majority-minority districts), and compactness (Polsby-Popper ratio).

The value of this analysis extends beyond academic interest. Structural gerrymandering elimination distinguishes this approach from boundary optimization methods [? ?]. Even if MMD boundaries are drawn with partisan bias, proportional seat allocation within districts prevents packing and cracking strategies from delivering disproportional outcomes. The system becomes effectively ungerrymandered not through careful boundary design but through the proportional allocation mechanism itself. Additionally, proportional representation addresses Voting Rights Act compliance in a fundamentally different way than single-member majority-minority districts [?]. Rather than concentrating minority populations geographically to create majority-minority districts, proportional allocation enables minority representation through vote share: a district with 35% Black voters electing 5 representatives would typically elect 2 Black representatives under cohesive voting patterns, without requiring geographic concentration that critics argue constitutes ”packing” [?].

Our contributions are threefold. First, we provide the first algorithmic design of multi-member districts for the U.S. House using recursive bisection, extending prior work on single-member algorithmic redistricting [? ?] to the MMD context. Second, we quantify the proportionality-compactness trade-off: larger multi-member districts (5 representatives) achieve near-perfect proportionality (Gallagher index < 2.0) but at the cost of reduced compactness (approximately 10% loss in Polsby-Popper ratio). Third, we demonstrate that 5-member MMDs enable minor party representation—parties receiving 3-5% of the vote within a district can win seats—while 3-member MMDs remain effectively two-party systems due to high exclusion thresholds [?].

2 Related Work

3 Methodology

4 Methodology

Our analysis proceeds in three stages: (1) generating multi-member district configurations via algorithmic redistricting, (2) simulating electoral outcomes using the D’Hondt proportional allocation method, and (3) measuring proportionality and compactness across config-

urations.

4.1 Geographic and Demographic Data

We use 2020 Census tract-level data for the contiguous United States (48 states plus D.C.), comprising 84,414 census tracts with a total population of 328.2 million. Tract boundaries and populations are obtained from the Census Bureau’s TIGER/Line shapefiles and PL 94-171 redistricting files [?]. For election data, we use 2020 presidential election results at the census tract level, obtained by disaggregating precinct-level returns from the MIT Election Data Science Lab [?] to census tracts using population-proportional allocation.

4.2 Multi-Member District Generation

We generate multi-member districts using edge-weighted recursive bisection graph partitioning, implemented in the `redist` open-source redistricting platform [?]. The algorithm represents census tracts as nodes in a graph, with edges weighted by shared boundary length. Graph partitioning minimizes total weighted edge cuts, equivalent to minimizing total district perimeter — producing compact districts without explicitly optimizing for a compactness metric.

We generate three uniform configurations:

- **Uniform-3:** 144 districts, 3 seats each (432 total seats)
- **Uniform-5:** 87 districts, 5 seats each (435 total seats)
- **Uniform-7:** 62 districts, 7 seats each (434 total seats)

using the `redist states --seats-per-district M --districts k` command with seed 42 for reproducibility.

4.2.1 Per-Seat Balance Criterion

For multi-member districts, balance is measured *per seat*, not per district. For a k -district configuration with M seats per district and total population P :

$$\text{ideal per seat} = \frac{P}{k \cdot M}, \quad \text{ideal per district} = M \cdot \text{ideal per seat} = \frac{P}{k}. \quad (1)$$

A district is balanced if its population deviates from the district ideal by no more than tolerance T . We target $T = 10\%$ for multi-member configurations, consistent with typical state legislative standards.

4.2.2 Per-Node Ufactor Scaling

A critical methodological contribution is the derivation of per-node METIS imbalance tolerances. Applying a fixed ufactor at each bisection level causes *compounding balance errors*: with D rounds at per-round tolerance ε , the worst-case cumulative deviation is $\varepsilon \cdot D$. For 7 rounds at $\varepsilon = 5\%$, the cumulative worst case is 35%.

We avoid this by computing the ufactor at each bisection node from its district count k_{node} and target tolerance T :

$$\text{ufactor}_{\text{node}} = 1 + \frac{T}{k_{\text{node}}}. \quad (2)$$

The root node of a 144-district map receives $\text{ufactor} = 1 + 0.10/144 \approx 1.00069$ (very tight); a leaf node splitting 2 final districts receives $\text{ufactor} = 1.05$ (5%). This guarantees cumulative balance error $\leq T$ regardless of bisection depth — a mathematical guarantee not achievable with fixed per-level ufactored.

Sensitivity to ufactor formulation. An earlier implementation passed ufactor values as floating-point strings to METIS, which uses `atoi()` for parsing: the fractional part was silently truncated to integer 1 (0.1% tolerance) regardless of the intended value. We verified that this implementation error does *not* materially affect the reported results because, at continental scale ($k \geq 62$ districts), the per-node ufactor formula yields $T/k_{\text{root}} \leq 0.16\%$ — which rounds to the same METIS integer value as the buggy implementation. The reported population deviations (± 3.7 – 4.3%) are dominated by the *granularity floor*: census block groups average 1,400 people, and one block-group swap represents $\sim 0.04\%$ of a multi-member district. METIS cannot achieve arbitrarily precise balance below this floor regardless of ufactor. The corrected per-node implementation provides a strict mathematical guarantee for smaller district counts ($k < 50$) where the floor does not bind.

4.3 Seat Allocation: D’Hondt Method

Within each multi-member district, seats are allocated using the D’Hondt highest-averages method [?]. For a district with M seats and parties receiving vote totals $\{v_i\}$, the algorithm awards seats iteratively: at each step, award the next seat to the party with the highest quotient $v_i/(s_i + 1)$, where s_i is that party’s current seat count. We implement this as the `dhondt_allocate` function in the `redist-analysis` crate, verified by unit tests including Ireland-style 3-party 5-seat constituencies.

The effective threshold — minimum vote share to win any seats — is approximately $1/(M + 1)$ (Droop quota): 25% for 3-member, 17% for 5-member, 12.5% for 7-member districts.

4.4 Proportionality Measurement

We measure proportionality using the Gallagher index [?]:

$$G = \sqrt{\frac{1}{2} \sum_i (v_i - s_i)^2} \quad (3)$$

where v_i and s_i are party i ’s vote and seat shares (in percentage points). A Gallagher index of 0 indicates perfect proportionality; typical FPTP systems score 10–15.

4.5 Compactness Measurement

We measure district compactness using the Polsby-Popper ratio [1]:

$$PP = \frac{4\pi A}{P^2} \quad (4)$$

where A is district area and P is perimeter. Scores range from 0 to 1 (perfect circle). We compute PP for dissolved district polygons using the `redist-analysis` compactness module on WGS84 coordinates.

4.6 Baseline Comparison

We compare multi-member configurations against: (1) the 2020 enacted single-member congressional districts, and (2) our edge-weighted single-member redistricting from Paper B.2 [?]. This isolates the effects of district magnitude from boundary-drawing method.

4.7 Replication

All code, data, and results are at <https://giodl73-repo.github.io/REDIST/>. The `redist` binary (version 0.1.0) reproduces all configurations from reported seeds. Each configuration’s manifest records the SHA-256 hash of source data and the exact command:

```
redist state --year 2020 --version E1 \  
  --districts [N] --seats-per-district [M] \  
  --balance-tolerance 10.0 --seed 42
```

5 Evaluation

6 Discussion

7 Conclusion

References

- [1] Daniel D Polsby and Robert D Popper. The third criterion: Compactness as a procedural safeguard against partisan gerrymandering. *Yale Law & Policy Review*, 9:301–353, 1991.
- [2] Nicholas Stephanopoulos and Eric McGhee. Partisan gerrymandering and the efficiency gap. *University of Chicago Law Review*, 2015.